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Tutorial Week 11

COMP3670 / COMP6670: Introduction to Machine Learning

The following three questions consolidate the mathematical ideas behind SVM, PCA and SVD. Focus on explaining why the conclusions hold, rather than only calculating the final answers.

Question 1 Proving that an SVM separator is optimal

A binary classification dataset contains six datapoints in $\mathbb{R}^2$:

i	1	2	3	4	5	6
$\mathbf{x}_i$	$\begin{pmatrix}6\\8\end{pmatrix}$	$\begin{pmatrix}10\\5\end{pmatrix}$	$\begin{pmatrix}12\\16\end{pmatrix}$	$\begin{pmatrix}-6\\-8\end{pmatrix}$	$\begin{pmatrix}-10\\-5\end{pmatrix}$	$\begin{pmatrix}-12\\-16\end{pmatrix}$
y_i	$+1$	$+1$	$+1$	-1	-1	-1

Recall that the hard-margin SVM solves

$$\begin{aligned} &\text{minimize} && \frac{1}{2}\|\mathbf{w}\|_2^2 \\ &\text{subject to} && y_i\big(\langle\ \mathbf{w}\ ,\ \mathbf{x}_i\ \rangle + b\big)\ \geq\ 1 \qquad (i = 1, \ldots, 6). \end{aligned}$$

We will find an optimal solution and prove its optimality without using an optimization solver.

(a) Consider $\mathbf{w} = t\begin{pmatrix}3\\4\end{pmatrix}$ and $b = 0$, where $t > 0$. Find the smallest t for which all six constraints hold, then calculate all six functional margins.

(b) For any feasible $\mathbf{w}, b$, use the constraints for $\mathbf{x}_1$ and $\mathbf{x}_4$ to show that

$$\langle\ \mathbf{w}\ ,\ \mathbf{x}_1 - \mathbf{x}_4\ \rangle\ \geq\ 2.$$

Apply the Cauchy-Schwarz inequality to obtain a lower bound on $\|\mathbf{w}\|_2$. Use this bound to prove that your solution from part (a) is globally optimal, even when b is allowed to vary.

Solution.

(a) The six functional margins are $50t, 50t, 100t, 50t, 50t, 100t$. Thus the smallest feasible value is $t = 1/50$, giving

$$\mathbf{w}_* \ =\ \frac{1}{50}\begin{pmatrix}3\\4\end{pmatrix}, \qquad b_* \ =\ 0.$$

The functional margins are $1, 1, 2, 1, 1, 2$.

(b) The two constraints give

$$\langle\ \mathbf{w}\ ,\ \mathbf{x}_1\ \rangle + b\ \geq\ 1, \qquad -\langle\ \mathbf{w}\ ,\ \mathbf{x}_4\ \rangle - b\ \geq\ 1.$$

Adding them eliminates b. Since $\mathbf{x}_1 - \mathbf{x}_4 = \begin{pmatrix} 12 \\ 16 \end{pmatrix}$ has norm 20, Cauchy-Schwarz gives

$$2 \;\le\; \langle\, \mathbf{w}\,,\; \mathbf{x}_1 - \mathbf{x}_4 \,\rangle \;\le\; \|\mathbf{w}\|_2 \, \|\mathbf{x}_1 - \mathbf{x}_4\|_2 \;=\; 20 \, \|\mathbf{w}\|_2 \,.$$

Hence every feasible solution has $\|\mathbf{w}\|_2 \ge 1/10$ and objective at least $1/200$. The feasible vector $\mathbf{w}_*$ has norm $5/50 = 1/10$, so it attains this lower bound and is globally optimal.

Question 2 **An orthonormal change of basis preserves total variance**

Assume the dataset $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$ is already centered, so $\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i = \mathbf{0}$. Let $\mathbf{q}_1, \ldots, \mathbf{q}_d$ be any orthonormal basis of $\mathbb{R}^d$ and set

$$\mathbf{Q} \;=\; \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_d \\ | & & | \end{pmatrix}, \qquad \mathbf{z}_i \;=\; \mathbf{Q}^{\mathsf{T}} \mathbf{x}_i .$$

Thus $\mathbf{z}_i$ gives all d coordinates of the datapoint in the new basis; no direction has yet been discarded. The *total variance* is the sum of the coordinate variances. Using the lecture's $1/n$ convention, the original centered dataset has total variance

$$T_X \;=\; \sum_{\ell=1}^{d} \frac{1}{n} \sum_{i=1}^{n} x_{i\ell}^2 .$$

Write T_Z for the total variance of the transformed dataset.

(a) Show that the transformed dataset is also centered and that

$$T_X \;=\; \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{x}_i\|_2^2 \,, \qquad T_Z \;=\; \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{z}_i\|_2^2 \,.$$

(b) Prove that $T_Z = T_X$. Your argument should work for any orthonormal basis, not just the principal-component directions. Does this imply that each new coordinate has the same variance as the corresponding original coordinate? Explain.

Solution.

(a) The original dataset is centered, so

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{z}_i \;=\; \mathbf{Q}^{\mathsf{T}} \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i \right) \;=\; \mathbf{0}.$$

Interchanging the finite sums gives

$$T_X \;=\; \sum_{\ell=1}^{d} \frac{1}{n} \sum_{i=1}^{n} x_{i\ell}^2 \;=\; \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{x}_i\|_2^2 \,.$$

The same calculation gives $T_Z \;=\; \frac{1}{n}\sum_{i=1}^n \|\mathbf{z}_i\|_2^2$, since the transformed dataset is also centered.

(b) Since $\mathbf{Q}$ is square with orthonormal columns, $\mathbf{Q}\mathbf{Q}^\mathsf{T} = \mathbf{I}_d$. For each datapoint,

$$\|\mathbf{z}_i\|_2^2 \;=\; \mathbf{x}_i^\mathsf{T}\mathbf{Q}\mathbf{Q}^\mathsf{T}\mathbf{x}_i \;=\; \|\mathbf{x}_i\|_2^2 .$$

The transformed mean is zero, so summing coordinate variances and averaging over datapoints yields

$$T_Z \;=\; \frac{1}{n}\sum_{i=1}^{n}\|\mathbf{z}_i\|_2^2 \;=\; \frac{1}{n}\sum_{i=1}^{n}\|\mathbf{x}_i\|_2^2 \;=\; T_X .$$

Individual coordinate variances need not be preserved. For example, take datapoints $(6,8)^\mathsf{T}$ and $(-6,-8)^\mathsf{T}$, whose coordinate variances are 36 and 64. The orthonormal basis

$$\mathbf{q}_1 \;=\; \frac{1}{5}\begin{pmatrix}3\\4\end{pmatrix}, \qquad \mathbf{q}_2 \;=\; \frac{1}{5}\begin{pmatrix}-4\\3\end{pmatrix}$$

gives transformed datapoints $(10,0)^\mathsf{T}$ and $(-10,0)^\mathsf{T}$, whose variances are 100 and 0. Both totals are 100.

Question 3 Why discarded singular values determine the approximation error

Let $\mathbf{A} \in \mathbb{R}^{m\times d}$ have rank $r \geq 1$ and compact SVD

$$\mathbf{A} \;=\; \sum_{\ell=1}^{r}\sigma_\ell \mathbf{u}_\ell \mathbf{v}_\ell^\mathsf{T}, \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0,$$

where the $\mathbf{u}_\ell$ and $\mathbf{v}_\ell$ form two orthonormal lists. Write $\mathbf{S}_\ell = \mathbf{u}_\ell\mathbf{v}_\ell^\mathsf{T}$, so $(\mathbf{S}_\ell)_{ab} = u_{a\ell}v_{b\ell}$. Recall that $\|\mathbf{M}\|_F^2 = \sum_a\sum_b M_{ab}^2$.

(a) Prove that

$$\sum_{a=1}^{m}\sum_{b=1}^{d}(\mathbf{S}_i)_{ab}(\mathbf{S}_j)_{ab} \;=\; \begin{cases}1, & i=j,\\ 0, & i\neq j.\end{cases}$$

Hint: Separate the sum involving the entries of the left singular vectors from the sum involving the right singular vectors.

(b) For $1 \leq k \leq r$, let $\mathbf{A}_k = \sum_{\ell=1}^{k}\sigma_\ell\mathbf{S}_\ell$. Expand the squares and use part (a) to prove

$$\|\mathbf{A}\|_F^2 \;=\; \sum_{\ell=1}^{r}\sigma_\ell^2, \qquad \|\mathbf{A}-\mathbf{A}_k\|_F^2 \;=\; \sum_{\ell=k+1}^{r}\sigma_\ell^2 .$$

You may use the Eckart-Young theorem to conclude that no matrix of rank at most k has smaller total squared reconstruction error; you need not prove its optimality inequality.

Solution.

(a) Using the entries of the outer products gives

$$\begin{aligned}\sum_{a=1}^{m}\sum_{b=1}^{d}(\mathbf{S}_i)_{ab}(\mathbf{S}_j)_{ab} \;&=\; \sum_{a=1}^{m}\sum_{b=1}^{d}u_{ai}v_{bi}u_{aj}v_{bj}\\ &=\; \left(\sum_{a=1}^{m}u_{ai}u_{aj}\right)\left(\sum_{b=1}^{d}v_{bi}v_{bj}\right)\\ &=\; \langle\, \mathbf{u}_i \,,\, \mathbf{u}_j \,\rangle\, \langle\, \mathbf{v}_i \,,\, \mathbf{v}_j \,\rangle .\end{aligned}$$

Both dot products are 1 when $i=j$ and 0 otherwise, proving the identity.

(b) For any set of indices $I \subseteq \{1, \dots, r\}$, expanding the square and interchanging finite sums gives

$$\begin{aligned} \left\| \sum_{\ell \in I} \sigma_\ell \mathbf{S}_\ell \right\|_F^2 &= \sum_{a=1}^{m} \sum_{b=1}^{d} \left(\sum_{\ell \in I} \sigma_\ell (\mathbf{S}_\ell)_{ab} \right)^2 \\ &= \sum_{i \in I} \sum_{j \in I} \sigma_i \sigma_j \left(\sum_{a=1}^{m} \sum_{b=1}^{d} (\mathbf{S}_i)_{ab} (\mathbf{S}_j)_{ab} \right) \\ &= \sum_{\ell \in I} \sigma_\ell^2 . \end{aligned}$$

The cross terms vanish by part (a). Choosing $I = \{1, \dots, r\}$ gives the first identity; choosing $I = \{k+1, \dots, r\}$ gives the residual identity. When $k = r$, the residual and the empty sum are both zero. By the Eckart-Young theorem, every matrix $\mathbf{B} \in \mathbb{R}^{m \times d}$ of rank at most k satisfies

$$\|\mathbf{A} - \mathbf{B}\|_F^2 \;\geq\; \|\mathbf{A} - \mathbf{A}_k\|_F^2 \;=\; \sum_{\ell=k+1}^{r} \sigma_\ell^2 .$$

The **average rate of change** of a function f between x and $x + \Delta x$ is the slope of the secant through the two points of its graph, and the **derivative** of f at x is its limit as the interval shrinks to the point:

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \qquad \text{and} \qquad f'(x) \;=\; \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} .$$

For a small step Δx this gives the **linear approximation** $f(x + \Delta x) \approx f(x) + f'(x)\, \Delta x$.

Table and rules. Here c is a constant, log is the natural logarithm, and f and g are differentiable functions:

function	c	x^n	e^x	$\log x$
derivative	0	$n\, x^{n-1}$	e^x	$1/x$

function	$f + g$	$c\, f$	$f\, g$	$g\big(f(x)\big)$
derivative	$f' + g'$	$c\, f'$	$f'g + f\, g'$	$g'\big(f(x)\big)\, f'(x)$

Recall also that $\log(ab) = \log a + \log b$ and $\log(e^x) = x$.

Question 4 **The derivative of a quadratic function**

Throughout this question

$$f(x) \;=\; a\, x^2 + b\, x + c,$$

where a, b and c are fixed numbers and $a \neq 0$. The in-sample error of the running example of Lecture 11, the line through the origin fitted to four observations, is of this form:

$$E_{\text{in}}(\theta) \;=\; 7.5\, \theta^2 - 30\, \theta + 31.5, \qquad \text{that is,} \quad a = 7.5, \quad b = -30, \quad c = 31.5.$$

(a) For the running example, compute the average rate of change of E_{in} between $\theta = 1$ and $\theta = 1 + \Delta\theta$, for $\Delta\theta = 1$, $\Delta\theta = 0.1$ and $\Delta\theta = 0.01$. You may use $E_{\text{in}}(1) = 9$, $E_{\text{in}}(2) = 1.5$, $E_{\text{in}}(1.1) = 7.575$ and $E_{\text{in}}(1.01) = 8.85075$. Which number do the three values approach? What do its **sign** and its **size** say about the error when θ is increased slightly from 1?

(b) Using the **definition** of the derivative, show that the average rate of change of f between x and $x + \Delta x$ is

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \;=\; 2a\,x + b + a\,\Delta x,$$

and conclude that $f'(x) = 2a\,x + b$. Write down $E'_{\text{in}}(\theta)$ and check your answer to (a).

(c) Show that the average rate of change of f between x and $x + \Delta x$ is equal to the derivative of f at the **midpoint** of the interval:

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} \;=\; f'\Big(x + \frac{\Delta x}{2}\Big).$$

Draw a parabola to see that, for a parabola, the secant through two points is parallel to the tangent at the point halfway between them. Use this to obtain, without computing any value of E_{in}, the average rate of change of E_{in} over each of the intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, $[3, 4]$ and $[0, 4]$.

(d) Show that, for every x and every step Δx, large or small,

$$f(x + \Delta x) \;=\; f(x) + f'(x)\,\Delta x + a\,(\Delta x)^2.$$

So the error of the linear approximation of a quadratic function is exactly $a\,(\Delta x)^2$. What happens to this error when the step is halved? In the lecture, the linear approximation of E_{in} at $\theta = 1$ predicted 7.5 for $E_{\text{in}}(1.1)$, whose true value is 7.575, and -6 for $E_{\text{in}}(2)$, whose true value is 1.5. Explain both errors.

(e) Find the value $x^\star$ at which $f'(x^\star) = 0$. A value at which the derivative is zero is called a **critical point** of f. What is the critical point of E_{in}? Use (c) to show that $f(x^\star - r) = f(x^\star + r)$ for every r: the graph of a quadratic function is symmetric about its critical point. Check it on $E_{\text{in}}(0) = E_{\text{in}}(4) = 31.5$ and $E_{\text{in}}(1) = E_{\text{in}}(3) = 9$.

(f) **(Optional)** Take $x = x^\star$ in (d). What does the formula say about $f(x^\star + \Delta x)$ compared with $f(x^\star)$ when $a > 0$? What have you proved about the line $h(x) = 2x$ in the running example?

(g) **(Optional)** The table of derivatives states that x^n has derivative $n\,x^{n-1}$ for *any* power n. Verify it from the definition for $n = 3$ and for $n = -1$, that is, for x^3 and for $1/x$.

Solution.

(a) The three average rates of change are

$$\frac{1.5 - 9}{1} = \mathbf{-7.5}, \qquad \frac{7.575 - 9}{0.1} = \mathbf{-14.25}, \qquad \frac{8.85075 - 9}{0.01} = \mathbf{-14.925},$$

and they approach $\mathbf{-15}$.

Sign: negative, so the error **decreases** when θ is increased slightly from 1. *Size*: the error changes by about 15 times the change in θ. For instance, increasing θ by 0.01 lowers the error by about 0.15 (exactly, by 0.14925).

(b) Expanding the square,

$$f(x + \Delta x) - f(x) \;=\; a\big[(x + \Delta x)^2 - x^2\big] + b\,\Delta x \;=\; 2a\,x\,\Delta x + a\,(\Delta x)^2 + b\,\Delta x,$$

and dividing by Δx gives $2a\,x + b + a\,\Delta x$. As $\Delta x \to 0$ the last term vanishes, so $f'(x) = \mathbf{2a\,x + b}$.

For the running example the average rate of change is $15\,\theta - 30 + 7.5\,\Delta\theta$, and $E'_{\text{in}}(\theta) = \mathbf{15\,\theta - 30}$. At $\theta = 1$ the average rate is $-15 + 7.5\,\Delta\theta$, which gives -7.5, -14.25 and -14.925 for the three steps of (a), and $E'_{\text{in}}(1) = -15$.

(c) By (b), $f'\big(x + \frac{\Delta x}{2}\big) = 2a\big(x + \frac{\Delta x}{2}\big) + b = 2a\,x + b + a\,\Delta x$, which is the average rate of change found in (b).

The midpoints of the five intervals are 0.5, 1.5, 2.5, 3.5 and 2, so the average rates of change are

$$E'_{\text{in}}(0.5) = \mathbf{-22.5}, \quad E'_{\text{in}}(1.5) = \mathbf{-7.5}, \quad E'_{\text{in}}(2.5) = \mathbf{7.5}, \quad E'_{\text{in}}(3.5) = \mathbf{22.5}, \quad E'_{\text{in}}(2) = \mathbf{0},$$

the table of the lecture.

(d) Expanding,

$$\begin{aligned} f(x + \Delta x) &= a\,x^2 + 2a\,x\,\Delta x + a\,(\Delta x)^2 + b\,x + b\,\Delta x + c \\ &= \underbrace{a\,x^2 + b\,x + c}_{f(x)} + \underbrace{(2a\,x + b)}_{f'(x)}\,\Delta x \;+\; a\,(\Delta x)^2. \end{aligned}$$

The error $a\,(\Delta x)^2$ does not depend on x, and halving the step divides it by **four**.

For the running example $a = 7.5$. A step of 0.1 gives an error of $7.5 \times 0.01 = \mathbf{0.075}$, which is $7.575 - 7.5$. A step of 1 gives an error of $7.5 \times 1 = \mathbf{7.5}$, which is $1.5 - (-6)$.

(e) $2a\,x^\star + b = 0$ gives $x^\star = \mathbf{-b/(2a)}$, the only critical point. For the running example, $\theta^\star = 30/15 = \mathbf{2}$.

The interval from $x^\star - r$ to $x^\star + r$ has midpoint $x^\star$, so by (c) the average rate of change of f over it is $f'(x^\star) = 0$. Hence $f(x^\star + r) - f(x^\star - r) = 0$. In the running example, 0 and 4 are both at distance 2 from the critical point, and 1 and 3 are both at distance 1.

(f) Since $f'(x^\star) = 0$, (d) reads $f(x^\star + \Delta x) = f(x^\star) + a\,(\Delta x)^2$. When $a > 0$ the last term is positive for every $\Delta x \neq 0$, so f **takes its smallest value at its critical point**, and nowhere else.

For the running example, $E_{\text{in}}(\theta) = 1.5 + 7.5\,(\theta - 2)^2$: among all the lines through the origin, $h(x) = 2x$ has the smallest in-sample error, namely 1.5.

(g) For x^3: $(x + \Delta x)^3 - x^3 = 3x^2\Delta x + 3x\,(\Delta x)^2 + (\Delta x)^3$, so the average rate of change is $3x^2 + 3x\,\Delta x + (\Delta x)^2$, which approaches $\mathbf{3x^2}$.

For $1/x$, with $x \neq 0$:

$$\frac{1}{\Delta x}\Big(\frac{1}{x + \Delta x} - \frac{1}{x}\Big) \;=\; \frac{1}{\Delta x} \cdot \frac{x - (x + \Delta x)}{x\,(x + \Delta x)} \;=\; -\frac{1}{x\,(x + \Delta x)},$$

which approaches $\mathbf{-1/x^2} = (-1)\,x^{-2}$.

Question 5 **(Optional) Critical points of functions used in this course**

Several results of this course were stated without derivation. Each of them comes down to finding the **critical points** of a function, that is, the values at which its derivative is zero, as we will see by the end of Lecture 11. In every part below: **(1)** compute the derivative, using the table and the rules above, and **(2)** find all the critical points.

(a) **The sample mean.** Let $y_1, \ldots, y_N$ be fixed numbers, with sample mean $\bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i$, and consider the average squared distance from a number c to them,

$$f(c) \;=\; \frac{1}{N}\sum_{i=1}^{N} \big(y_i - c\big)^2.$$

Compute $f'(c)$ and find the critical points.

(b) **The impurity of a region.** For a region in which a proportion $p \in (0,1)$ of the points belongs to class 1, Lecture 4 defined the Gini impurity and the entropy,

$$G(p) \;=\; 2\,p\,(1-p) \qquad \text{and} \qquad H(p) \;=\; -p\log p - (1-p)\log(1-p),$$

and stated that both peak at $p = \frac{1}{2}$. Compute $G'(p)$ and $H'(p)$ and find the critical points of each.

(c) **Maximum likelihood for the Normal distribution.** Let $Z_1, \ldots, Z_N$ be an i.i.d. sample from the Normal distribution with mean μ and variance $v = \sigma^2$, whose density is $p(z \mid \mu, v) = \frac{1}{\sqrt{2\pi v}}\, e^{-(z-\mu)^2/(2v)}$. Minus the log-likelihood of the sample is

$$-\mathrm{LL}(\mu, v) \;=\; \frac{N}{2}\log(2\pi) + \frac{N}{2}\log v + \frac{1}{2v}\sum_{i=1}^{N}\big(Z_i - \mu\big)^2.$$

(i) Treat v as a fixed number, so that $-\mathrm{LL}$ is a function of μ only. Compute its derivative and find the critical point $\hat{\mu}$. Compare with (a).

(ii) Now treat μ as a fixed number, so that $-\mathrm{LL}$ is a function of v only. Compute its derivative and find the critical point $\hat{v}$. [*Hint*: $1/v = v^{-1}$.]

(iii) Which pair $(\hat{\mu}, \hat{v})$ makes both derivatives zero at the same time? What can you conclude about the maximum likelihood estimator of the mean and variance of the Normal distribution?

(iv) Obtain the expression of $-\mathrm{LL}(\mu, v)$ from the density.

(d) **The logistic function and the cross-entropy loss.** Let $\sigma(z) = \dfrac{1}{1+e^{-z}}$.

(i) The lecture obtained $\sigma'(z) = \dfrac{e^{-z}}{(1+e^{-z})^2}$ with the chain rule, and asked why this equals $\sigma(z)\big(1-\sigma(z)\big)$. Show it. Does σ have critical points?

(ii) Take one observation (x, y), where $x \neq 0$ is a number and $y \in \{0, 1\}$, and a model with a single weight w that predicts the probability $\sigma(w\,x)$. Its cross-entropy loss is

$$\ell(w) \;=\; -\,y\log\sigma(w\,x) - (1-y)\log\big(1-\sigma(w\,x)\big).$$

Show that $\ell'(w) = \big(\sigma(w\,x) - y\big)\,x$. Compare with the derivative of the squared loss obtained in the lecture, $-2\,x\,(y - \theta x)$.

(iii) Does ℓ have a critical point? For a sample $(x_1, y_1), \ldots, (x_N, y_N)$, write the equation that a critical point of the average loss $\frac{1}{N}\sum_{i=1}^{N}\ell_i(w)$ must satisfy. Can you solve it for w?

Solution.

(a) By the chain rule, with inside $u = y_i - c$ (derivative -1) and outside u^2 (derivative $2u$), each term has derivative $-2\,(y_i - c)$. By the sum and constant multiple rules,

$$f'(c) \;=\; -\frac{2}{N}\sum_{i=1}^{N}\,(y_i - c) \;=\; -2\,(\bar{y} - c) \;=\; \boldsymbol{2\,(c - \bar{y})},$$

which is zero only at $c = \boldsymbol{\bar{y}}$: the sample mean is the only critical point.

(b) *Gini.* By the product rule, $G'(p) = 2\big[1 \times (1-p) + p \times (-1)\big] = \boldsymbol{2 - 4p}$, which is zero only at $p = \boldsymbol{1/2}$.

Entropy. By the product rule, $p \log p$ has derivative $\log p + p \times \frac{1}{p} = \log p + 1$. By the product and chain rules, $(1-p)\log(1-p)$ has derivative $-\log(1-p) + (1-p) \times \frac{1}{1-p} \times (-1) = -\log(1-p) - 1$. Hence

$$H'(p) \;=\; -\big(\log p + 1\big) + \big(\log(1-p) + 1\big) \;=\; \boldsymbol{\log(1-p) - \log p},$$

which is zero when $\log(1-p) = \log p$, that is, only at $p = \boldsymbol{1/2}$.

Both impurities have a single critical point, the evenly split region.

(c) (i) The first two terms do not involve μ, and each term of the sum has derivative $-2\,(Z_i - \mu)$, as in (a). So the derivative in μ is

$$-\frac{1}{v}\sum_{i=1}^{N}\,(Z_i - \mu) \;=\; \frac{N}{v}\,(\mu - \bar{Z}),$$

which is zero only at $\hat{\mu} = \boldsymbol{\bar{Z}}$, whatever the value of v. It is the computation of (a): for the mean of a Normal distribution, maximum likelihood and least squares lead to the same critical point.

(ii) Write $S = \sum_i (Z_i - \mu)^2$, a fixed number. Since $\log v$ has derivative $1/v$ and $\frac{1}{2v} = \frac{1}{2}v^{-1}$ has derivative $-\frac{1}{2}v^{-2}$, the derivative in v is

$$\frac{N}{2v} - \frac{S}{2v^2} \;=\; \frac{1}{2v^2}\,(N\,v - S),$$

which is zero only at $\hat{v} = \boldsymbol{S/N} = \dfrac{1}{N}\sum_{i=1}^{N}(Z_i - \mu)^2$.

(iii) By (i) the mean must be $\bar{Z}$, and then by (ii)

$$\hat{\mu} \;=\; \frac{1}{N}\sum_{i=1}^{N} Z_i, \qquad\qquad \hat{\sigma}^2 \;=\; \hat{v} \;=\; \frac{1}{N}\sum_{i=1}^{N}\,\big(Z_i - \hat{\mu}\big)^2,$$

the **sample mean** and the **sample variance** of Lecture 4.

(iv) The likelihood is the product of the N densities, so the log-likelihood is the sum of their logarithms, and

$$\log p(Z_i \mid \mu, v) \;=\; -\frac{1}{2}\log(2\pi v) - \frac{(Z_i - \mu)^2}{2v} \;=\; -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log v - \frac{(Z_i - \mu)^2}{2v}.$$

Adding the N terms and changing the sign gives the expression.

(d) (i) $1-\sigma(z)=\dfrac{(1+e^{-z})-1}{1+e^{-z}}=\dfrac{e^{-z}}{1+e^{-z}}$, so

$$\sigma(z)\big(1-\sigma(z)\big) \;=\; \frac{1}{1+e^{-z}}\cdot\frac{e^{-z}}{1+e^{-z}} \;=\; \frac{e^{-z}}{(1+e^{-z})^2} \;=\; \sigma'(z).$$

σ has **no critical point**: $\sigma(z)$ and $1-\sigma(z)$ are both positive for every z, so $\sigma'(z)>0$.

(ii) Write $p=\sigma(w\,x)$. By the chain rule and (i), p has derivative $p\,(1-p)\,x$ in w. Then, by the chain rule again,

$$\frac{d}{dw}\big[-y\log p\big] \;=\; -\frac{y}{p}\,p\,(1-p)\,x \;=\; -y\,(1-p)\,x,$$
$$\frac{d}{dw}\big[-(1-y)\log(1-p)\big] \;=\; \frac{1-y}{1-p}\,p\,(1-p)\,x \;=\; (1-y)\,p\,x.$$

Adding, $\ell'(w)=\big[-y+y\,p+p-y\,p\big]\,x=\boldsymbol{(p-y)\,x}=\big(\sigma(w\,x)-y\big)\,x$.
The squared loss has derivative $2\,x\,(\theta x-y)$. In both cases the derivative is the **input times the difference between the prediction and the response**.

(iii) $\ell'(w)=0$ would need $\sigma(w\,x)=y$, with y equal to 0 or 1. But $0<\sigma(z)<1$ for every z, so ℓ has **no critical point**.
For a sample, by the sum and constant multiple rules, a critical point must satisfy

$$\frac{1}{N}\sum_{i=1}^{N}\big(\sigma(w\,x_i)-y_i\big)\,x_i \;=\; 0, \qquad \text{that is,} \qquad \sum_{i=1}^{N}\sigma(w\,x_i)\,x_i \;=\; \sum_{i=1}^{N}y_i\,x_i.$$

The unknown w sits inside σ in every term and cannot be isolated: there is **no formula** for the solution. This is the "no closed form" of Lecture 4, and the reason why logistic regression is fitted by an algorithm.